High-Level Design – TFA Settings

TaskFlow – Intelligent Task Allocation & Tracking System

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Based on: BRD-TFA-Settings v1.0, FRD-TFA-Settings v1.0
Module: Settings · **Sub-module:** 06.3 TFA Settings · **Code:** TFAS
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Document Control

Version	Date	Author	Summary of changes
1.0	31-Oct-2026	Jojo George	Initial version

Related documents:

Document	Location
Overall HLD (stack, deployment, cross-cutting design)	`../../00-Overall/HLD-TaskFlow-Overall.md`
BRD – TFA Settings	`../../../BRD/06-Settings/06.3-TFA-Settings/BRD-TFA-Settings.md`
FRD – TFA Settings	`../../../FRD/06-Settings/06.3-TFA-Settings/FRD-TFA-Settings.md`
LLD – TFA Settings	`../../../LLD/06-Settings/06.3-TFA-Settings/LLD-TFA-Settings.md`
HLD – TFA Assistant (runtime consumer)	`../../08-TFA-Assistant/08.1-Assistant/HLD-Assistant.md`
HLD – Role Configuration (permission prerequisites)	`../06.4-Role-Configuration/HLD-Role-Configuration.md`
TDD	`TF - Documents/TF - TDD/TDD-TaskFlow.md` §3.2 (key ring, AI providers), §6.5 (Integrations), §8.9 (Settings)

1. Introduction

1.1 Purpose

This HLD describes the architecture of TFA Settings, the screen and services that decide which AI provider and model the TFA Assistant uses: provider configuration, protected storage of API keys, the

model list, the connection test and the provider clients reached through the Assistant's model gateway. Class- and column-level detail is in the LLD.

1.2 Scope

In scope: the TFA Settings screen (`/settings/tfa`); its API under `/api/v1/assistant/providers` (read and save configuration, remove an added model, test a provider); the configuration service, connection tester, key protector, model and configuration storage, first-run seeding; the provider clients for Ollama, Google Gemini and OpenAI.

Out of scope: the TFA chat itself, tools, confirm-before-act and the assistant's provider selector (TFA Assistant HLD); editing permissions (Role Configuration HLD).

1.3 Audience

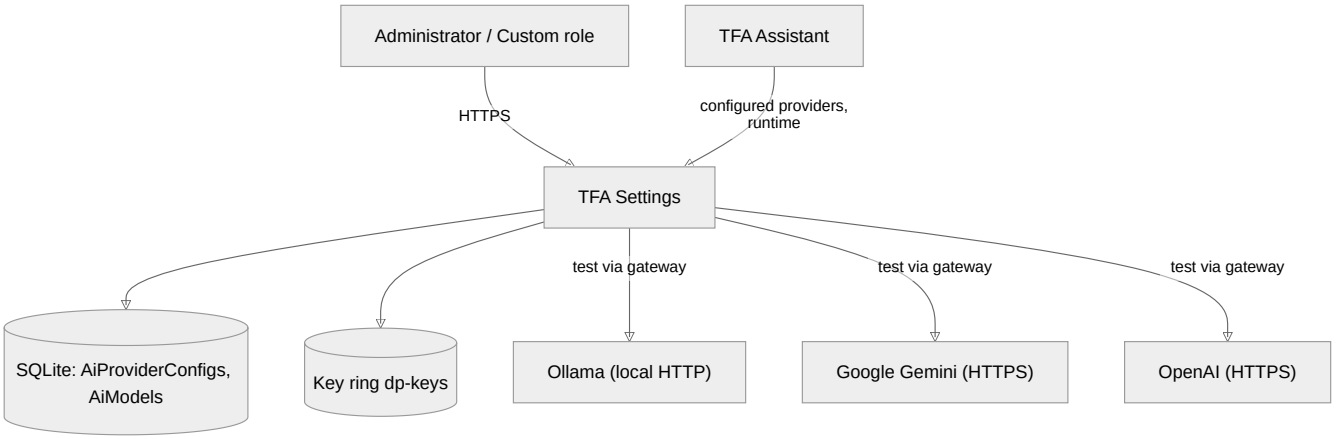
Developer, reviewer and tester of TaskFlow; the product owner for the design decisions.

1.4 Definitions & Abbreviations

Term	Meaning
TFA	TaskFlow Assistant, the AI chat assistant inside TaskFlow
Provider	An AI service: Ollama (local), Google Gemini or OpenAI (cloud)
Configured provider	Switched on, has a model, and has an address (Ollama) or a readable key (cloud)
Data Protection	ASP.NET Core encryption API; keys kept in a key ring folder
Key ring	Folder `dp-keys` (`AiProviders:KeyRingPath`) under the data folder; DPAPI-protected on Windows
Model gateway	`ChatModelGateway` (`IChatModelGateway`): the single path from TaskFlow to any provider
SSRF	Server-side request forgery – making the server call an address it should not

2. System Context

2.1 Context Diagram



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2.2 Actors & External Systems

Actor / system	Interaction	Protocol
Administrator; Custom role with `settings.ai.*`	View, edit, test, remove models	HTTPS / JSON
TFA Assistant (in process)	Reads which providers are configured and the runtime settings (model, address, decrypted key)	In-process service call
Ollama	Local model server, no key	HTTP to the saved server address
Google Gemini	Cloud model API, key in a request header	HTTPS / REST
OpenAI	Cloud model API, bearer key	HTTPS / REST
Data Protection key ring	Encrypts and decrypts API keys	Local files

3. Architecture Goals, Constraints & Principles

3.1 Quality Goals

Goal	How the architecture supports it
Secret safety	API keys encrypted with Data Protection before storage; write-only through the API (only "key set" and the last four characters are returned) (NFR-TFAS-01)
SSRF protection	The server-address validator refuses link-local and cloud-metadata hosts; provider clients do not follow redirects (NFR-TFAS-02)
Least privilege	Four fine-grained permissions (view, edit, test, remove model), each checked on the server (NFR-TFAS-03)
Robust test	The connection test always returns a result message, never an error status (NFR-TFAS-04)
Detect lost keys	Each read checks whether the stored key can still be decrypted (NFR-TFAS-05)
Plain language	Provider failures are classified (not configured, invalid key, rate limited, timeout, unreachable, model unavailable) and mapped to fixed messages (NFR-TFAS-06)
Privacy	Cloud providers are tagged; Ollama (local) is the default on a new installation (NFR-TFAS-07)

3.2 Constraints

- Exactly three providers, fixed in code (`AiProviderCatalog`).
- One API process; provider calls are synchronous HTTP with a timeout of `AiProviders:TimeoutSeconds` (60).
- Key protection depends on the key ring folder staying with the data folder; losing it makes saved keys unreadable.
- Only the screen enforces "one active server"; the API accepts several enabled providers.

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3.3 Architecture Principles

- Controller → Service → Repository; integrations (key protection, provider clients) behind interfaces (overall HLD).
- Every call to a model goes through the one model gateway – the test uses the same path as the assistant.
- Secrets never leave the server towards the browser and never appear in logs.
- Defence in depth: SPA hides buttons, the API checks each permission, the validator checks each value.

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4. Architecture Overview

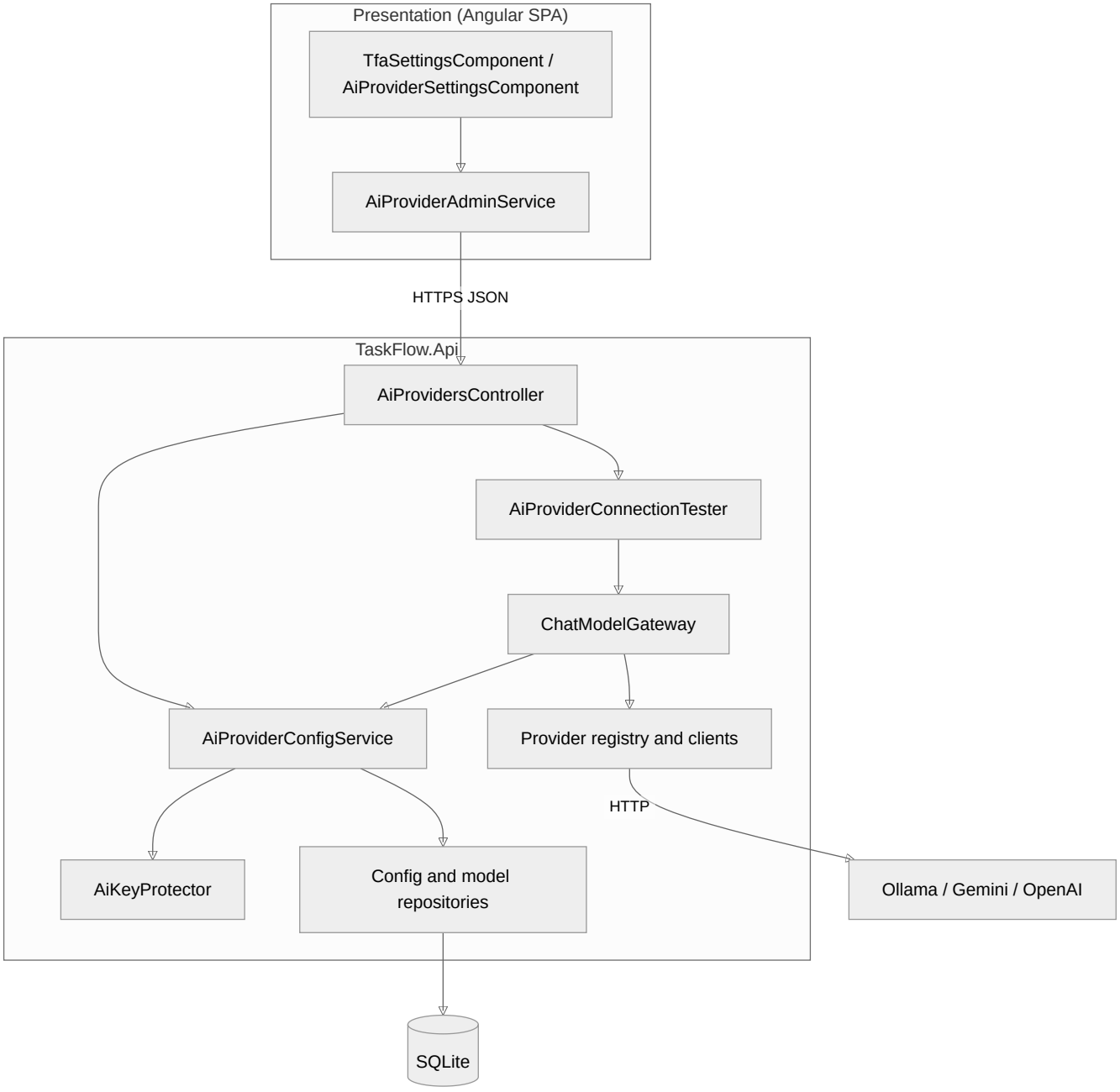
4.1 Architecture Style

An Angular screen calls a REST controller with one permission per action. Behind it, a configuration service owns the provider rows, the model list and key protection; a connection tester sends one short request through the Assistant's model gateway, which picks the provider client from a registry.

API surface (4 actions; full list in the LLD): `/api/v1/assistant/providers/config` (read, save), `/api/v1/assistant/providers/{kind}/models` (remove an added model), `/api/v1/assistant/providers/{kind}/test` (connection test). The assistant's own provider selector reads `/api/v1/assistant/providers`, owned by the TFA Assistant.

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4.2 Logical Architecture



4.3 Layers & Responsibilities

Layer	Responsibility in this sub-module	May call
Presentation	Provider card and status, model selector with built-in and added models, server address, API key box with keep/remove, Save, Test, Remove model; buttons shown per permission	API
API / Controllers	`AiProvidersController`: one policy per action; routes `{kind}` to the provider	Services
Validation	Request validators: providers present and unique, model name, server address (http/https, no link-local or metadata host), key without spaces, not new key and clear together	–

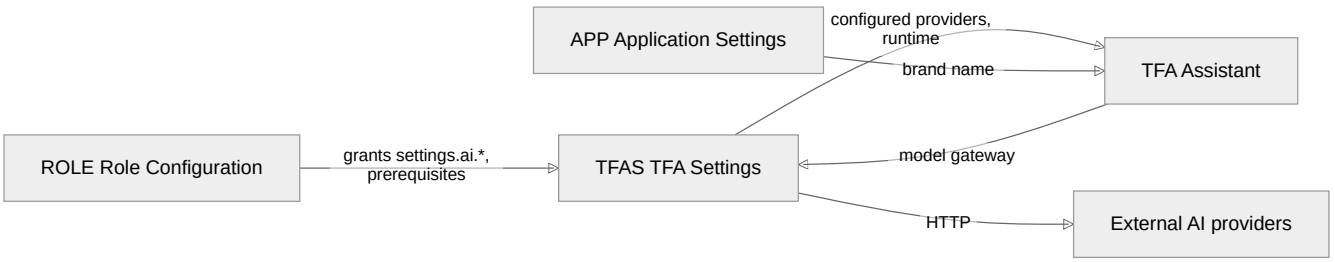
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Layer	Responsibility in this sub-module	May call
Services	`AiProviderConfigService` : read config with status, save, remember added models, remove model, runtime for the gateway; `AiProviderConnectionTester` : one test request and result mapping	Repositories, integrations
Repositories	`AiProviderConfigRepository` , `AiModelRepository`	Database
Integrations	`AiKeyProtector` (Data Protection), `KeyRingProtectionCheck` , provider registry and the Ollama, Gemini and OpenAI clients	Key ring, external providers
Start-up	`AiProviderSeeder` : built-in models and one row per provider	Database

5. Component / Module Design

Module / component	Responsibility	Key components (high level)	Depends on
TFA Settings screen	Choose server and model, enter address or key, save, test, remove added model	`TfaSettingsComponent` , `AiProviderSettingsComponent` , `AiProviderAdminService`	TFA Settings API, current user's permissions
TFA Settings API	Authorize and route the four actions	`AiProvidersController` , request validators	Config service, connection tester
Provider configuration	Rows per provider, status (configured, key readable), model list, save rules, runtime settings	`AiProviderConfigService` , `AiProviderCatalog`	Repositories, key protector
Connection test	One short request, plain result	`AiProviderConnectionTester` , `ProviderFailureMessages`	Model gateway
Key protection	Encrypt/decrypt keys; warn about an unprotected key ring at start-up	`AiKeyProtector` , `KeyRingProtectionCheck`	Data Protection key ring
Provider clients	Call each provider; classify failures; no redirects	`ChatModelProviderRegistry` , `OllamaProvider` , `GeminiProvider` , `OpenAiProvider`	External providers
Model gateway (Assistant module)	Single path to a provider for test and chat	`ChatModelGateway`	Config service (runtime), registry
Seeding	First-run defaults	`AiProviderSeeder`	SQLite

5.1 Module Interaction



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Interaction	What is exchanged
TFAS → TFA	Which providers are configured (assistant model selector) and the runtime settings for each call
TFA → TFAS	The connection test uses the Assistant's model gateway, so a passing test means chat will reach the provider
ROLE → TFAS	Grants of `settings.ai.view`, `settings.ai.edit`, `settings.ai.test`, `settings.ai.model.remove`; edit, test and remove each require view
TFAS → providers	One short chat request for a test; chat requests from the assistant use the same clients

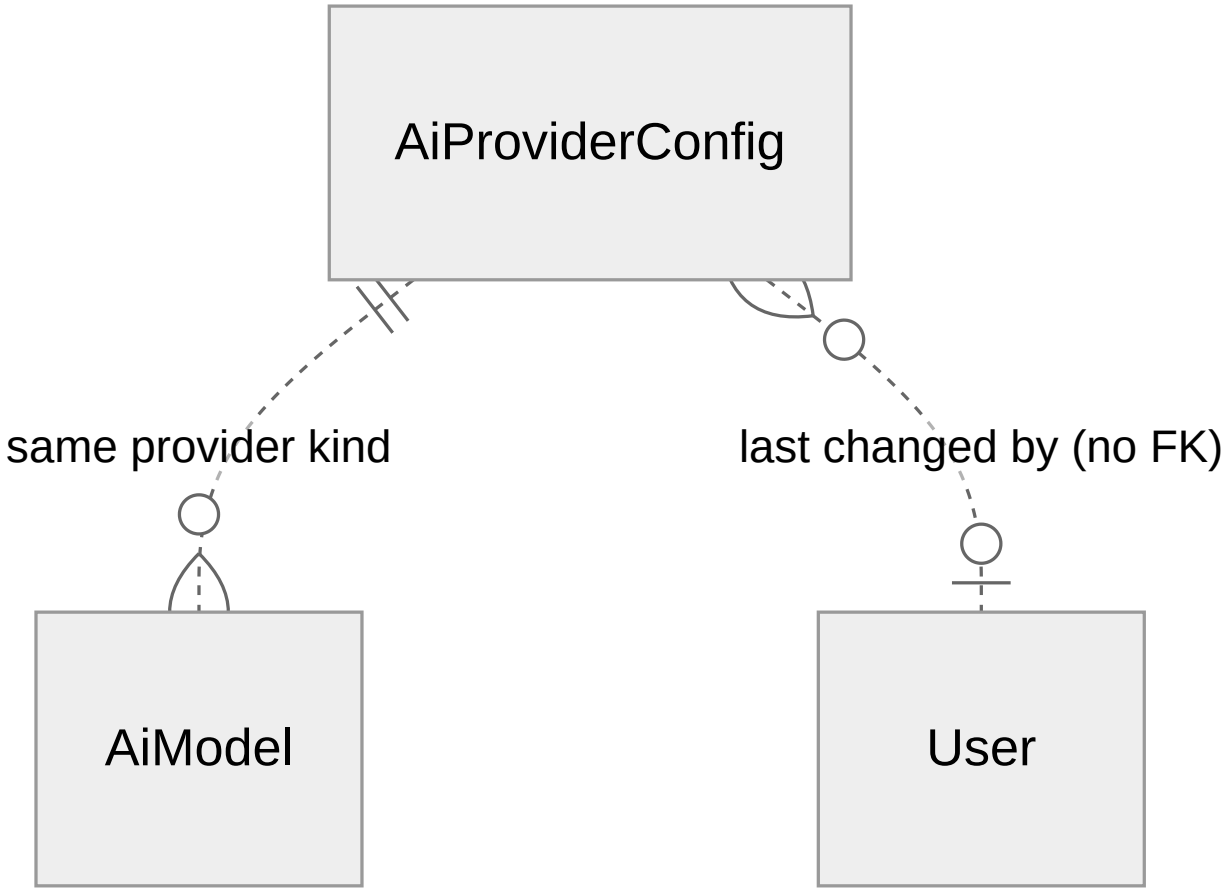
6. Data Architecture (high level)

6.1 Data Stores

Store	Technology	Holds	Notes
Main database	SQLite (`taskflow.db`, EF Core)	One configuration row per provider (on/off, model, address, encrypted key, last four characters, editor and time); the list of models per provider (built-in and added)	Encrypted key never returned
Key ring	Data Protection key files in `dp-keys`	Keys that encrypt the API keys (purpose `TaskFlow.AiProviderKeys`)	DPAPI-protected on Windows; checked at start-up; must be backed up with the database
Code catalogue	`AiProviderCatalog`	Providers, display names, built-in models, defaults	Fixed in code

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6.2 Main Data Domains



Domain	Owner	Use here
`AiProviderConfig`	TFAS	Read and written; runtime read by the TFA Assistant
`AiModel`	TFAS	Read; added models written on save and removed on request; built-in models seeded

6.3 Data Flow

1. First start seeds the built-in models and one row per provider; on a brand-new installation Ollama is switched on at `http://localhost:11434`, cloud providers stay off without a key (FR-TFAS-24).
2. The screen reads all providers with their status, model lists and "key set / last four" – never the key.
3. On save the SPA switches the chosen provider on and the others off; a new key is encrypted before storage; an empty key box keeps the stored key; a new model name is remembered in the list.
4. The test and the assistant ask the configuration service for the runtime settings; the key is decrypted only in memory for the outgoing request.

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7. Integration Design

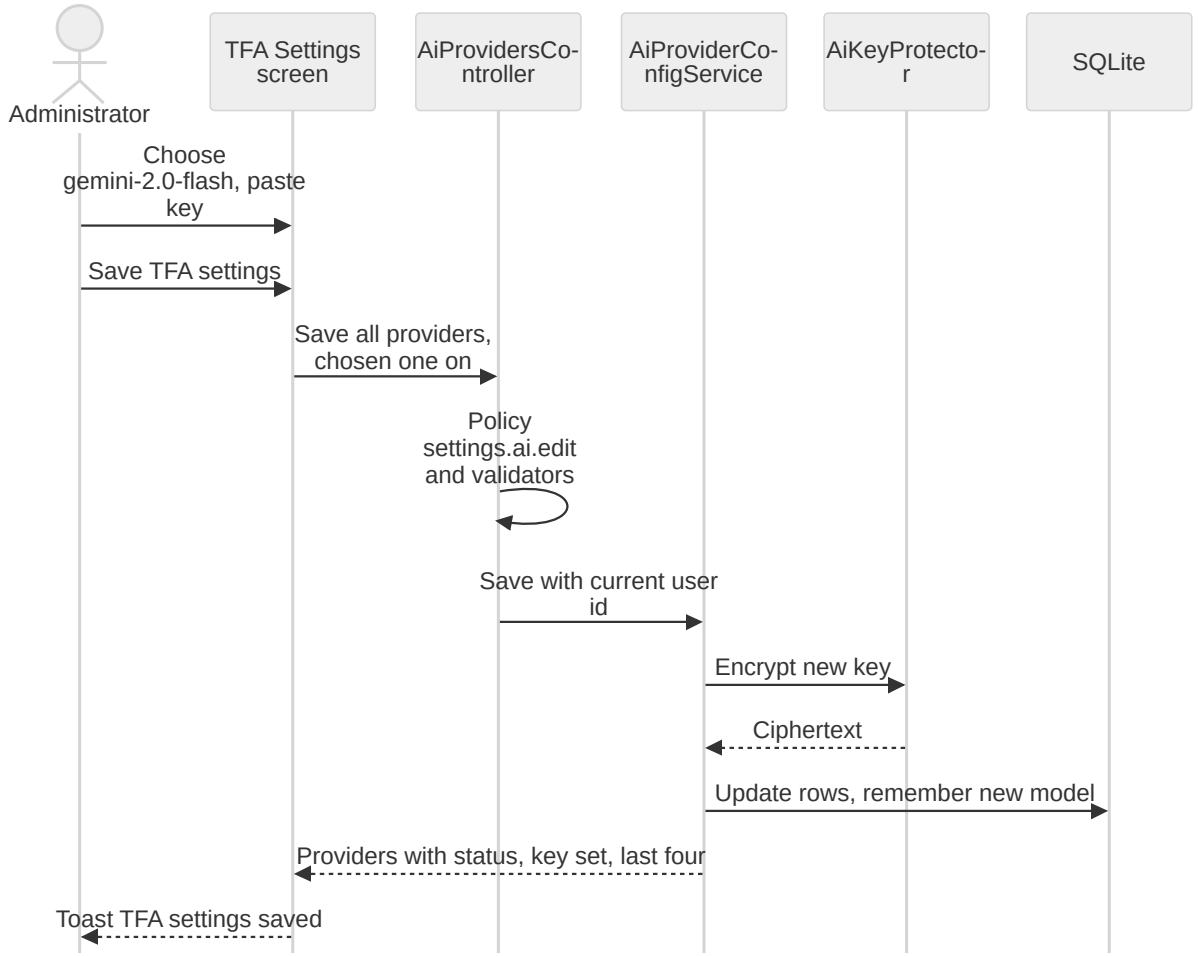
Integration	Direction	Purpose	Failure behaviour
Ollama (local model server)	Outbound HTTP to the saved address	Test and chat	Unreachable / timeout / model missing → fixed plain message; TFA not available until fixed
Google Gemini	Outbound HTTPS, key in a request header	Test and chat	Invalid key, rate limit (with retry time), timeout, model unavailable → fixed plain message
OpenAI	Outbound HTTPS, bearer key	Test and chat	As for Gemini
Data Protection key ring	Local	Encrypt/decrypt keys	Unreadable key → status "The saved key can no longer be read. Enter it again."; provider treated as not configured; a warning is logged
TFA Assistant	Inbound (in process)	Configured providers and runtime	Not-configured provider is not offered

All provider calls share a timeout (`AiProviders:TimeoutSeconds`), do not follow redirects, and log only provider, failure kind and HTTP status – no headers, bodies or keys. No automatic switch to another provider on failure.

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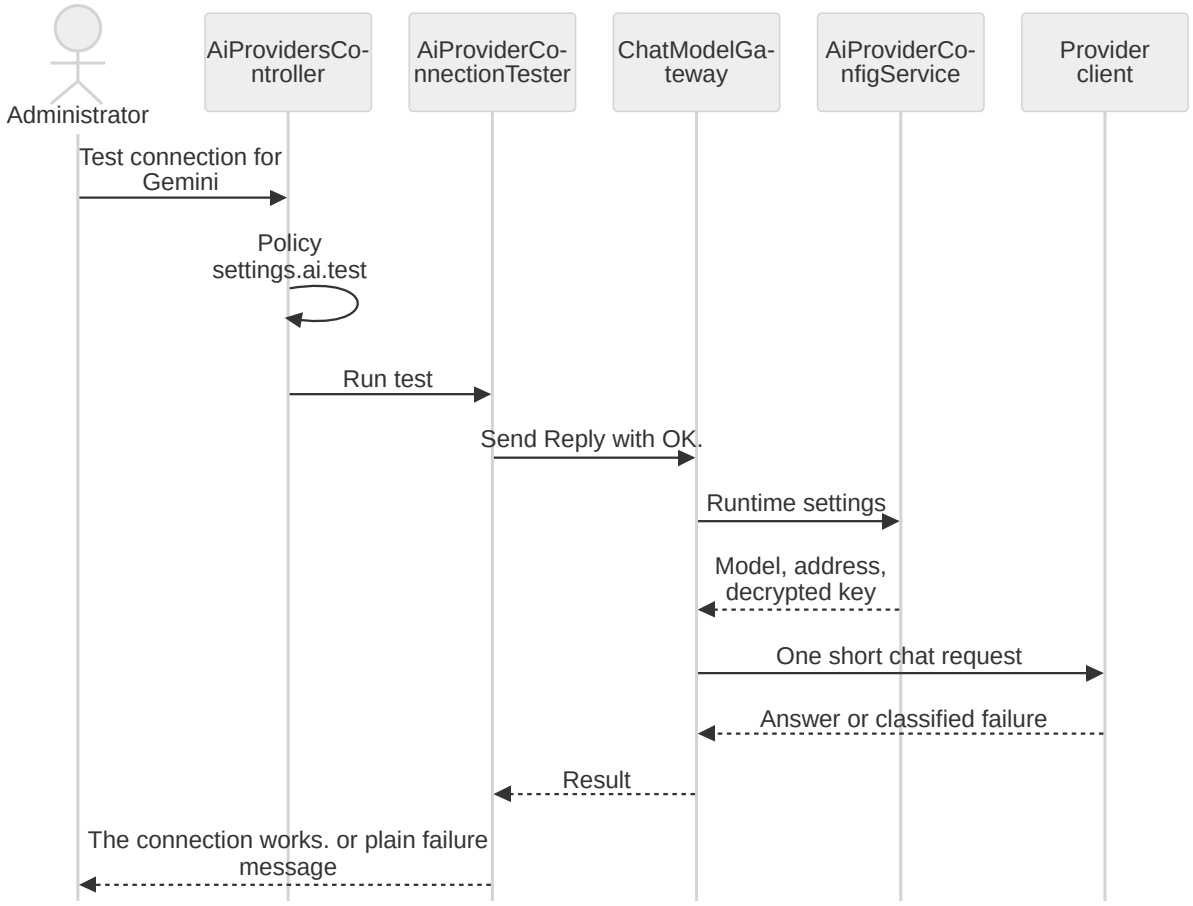
8. Key Flows (high level)

8.1 Choose provider and model, save



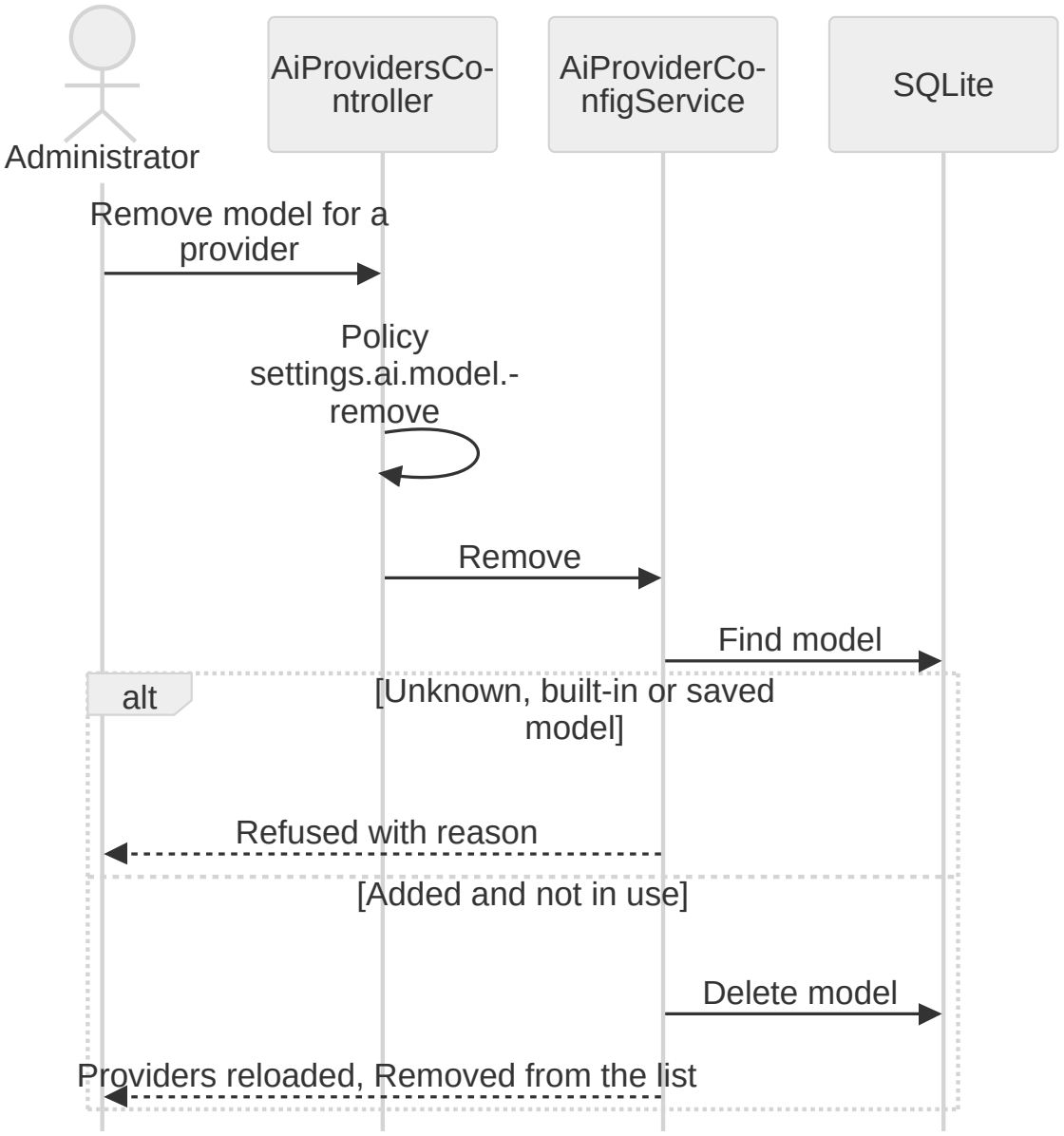
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8.2 Test the connection (integration)



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8.3 Remove an added model



9. Security Architecture

Concern	Approach
Authentication	JWT bearer access token and refresh cookie; session revalidated on every request (overall HLD)
Authorization	Controller requires sign-in; each action has its own policy: read `settings.ai.view`, save `settings.ai.edit`, test `settings.ai.test`, remove model `settings.ai.model.remove` (FR-TFAS-25). Edit, test and remove each require view, enforced by Role Configuration (FR-TFAS-26). The legacy key `settings.ai.manage` expands to the four keys for roles that still hold it. SPA: menu item and `permissionGuard` on view; `canEdit`, `canTest`, `mayRemoveModel` hide the buttons
Default grants	Administrator only; Manager and Employee denied; custom roles configurable

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Concern	Approach
Secret handling	Keys encrypted with Data Protection (purpose `TaskFlow.AiProviderKeys`); write-only API; only "key set" and last four characters returned; key ring DPAPI-protected on Windows and checked at start-up (`KeyRingProtectionCheck`)
SSRF	Server address must be http/https and not link-local or a cloud metadata host; `localhost` and private addresses allowed for Ollama; redirects not followed
Transport	HTTPS from the browser; HTTPS to cloud providers; Ollama over the configured address
Privacy	Screen explains that cloud providers receive the user's question and related task details (FR-TFAS-23)
Auditing	Last editor and time per provider row; denied requests logged by the denied-action audit handler; provider logs carry no secrets

10. Technology Stack

TFA Settings uses the shared TaskFlow stack: Angular 19 SPA, ASP.NET Core Web API on .NET 10, EF Core 10 with SQLite, FluentValidation 12. Specific to this sub-module are ASP.NET Core Data Protection for key encryption and typed `HttpClient` provider clients for Ollama, Google Gemini and OpenAI (no vendor SDK packages). Versions and rationale are in the overall HLD (`../../00-Overall/HLD-TaskFlow-Overall.md`, §10).

11. Deployment Architecture

TFA Settings runs inside the single `TaskFlow.Api` process. Deployment-relevant items: the key ring folder `AiProviders:KeyRingPath` (default `dp-keys`) under `Storage:Root` must stay with the database (and be backed up with it) or saved keys become unreadable; `AiProviders:TimeoutSeconds` (60) bounds each provider call; Ollama, when used, runs as a separate local service reachable at the saved address (default `http://localhost:11434`); cloud providers need outbound HTTPS. Environments, hosting and backup are in the overall HLD (`../../00-Overall/HLD-TaskFlow-Overall.md`, §11).

12. Non-Functional Design

Area	Design decision
Performance	Configuration read and save are small database operations; a test is one short provider request bounded by the timeout
Scalability	Three provider rows and a short model list; single instance
Reliability	Test never throws to the screen (always a result); unreadable key detected and reported; duplicate-model race on save handled by retrying without the duplicate
Availability	A provider outage affects only TFA; the rest of TaskFlow is independent
Security	See §9; secrets never logged or returned

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Area	Design decision
Maintainability	Provider list and built-in models in one catalogue; each provider behind one client interface in a registry; the test reuses the assistant's gateway
Observability	Provider failures logged by provider, failure kind and status; key-ring warning at start-up; warning when an enabled provider's key cannot be decrypted
Backup & Recovery	Database and key ring together; see overall HLD

13. Design Decisions

ID	Decision	Alternatives considered	Rationale
D-TFAS-1	Encrypt API keys with ASP.NET Core Data Protection and a key ring in the data folder	Plain text in the database; keys in `appsettings.json`; OS secret store	Keys not readable from the database file; no extra dependency; one folder to back up
D-TFAS-2	Keys are write-only: the API returns only "key set" and the last four characters	Return masked or full key	A key can never leak through the browser (BR-TFAS-01)
D-TFAS-3	Three fixed providers behind one model gateway and a provider registry	Vendor SDKs; a generic OpenAI-compatible endpoint only	Free local option (Ollama) plus two cloud options; one path for test and chat; easy to add a provider in code
D-TFAS-4	The connection test goes through the same gateway as the assistant and always returns 200 with a result	Separate ping endpoint per provider; error status on failure	A passing test proves chat will work; the screen shows a message, never an error page
D-TFAS-5	Fine-grained permissions view / edit / test / remove model, each requiring view	One `settings.ai.manage` key (now legacy)	Lets an organization delegate testing without editing; legacy key still honoured
D-TFAS-6	Server-address validator blocks link-local and metadata hosts; no redirects	Allow any URL	Prevents SSRF towards cloud metadata services (BR-TFAS-06)
D-TFAS-7	"One active server" applied by the screen; the API stores each provider's own on/off	Enforce a single enabled provider in the API	Switched-off providers keep their keys, models and addresses for a quick switch back
D-TFAS-8	Ollama on by default only on a brand-new installation	Always on; always off	Works out of the box locally without sending data to the cloud; upgrades are not changed silently
D-TFAS-9	Models are data (`AiModel`), built-in ones seeded and protected, added ones removable when not in use	Free-text model only	Keeps the selector tidy and prevents removing the model in use (BR-TFAS-04)

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14. Risks, Assumptions & Open Issues

Type	Description	Mitigation / owner
Risk	API key leaked (R-TFAS-01)	Encrypted at rest, write-only API, no logging – Jojo George
Risk	Task data sent to a cloud provider without awareness (R-TFAS-02)	Cloud tag and explanation; Ollama default – Jojo George
Risk	Cloud rate limit reached (R-TFAS-03)	Rate-limit message with retry time; switch model – Jojo George
Risk	Key ring lost or unprotected, keys unreadable (R-TFAS-04)	Start-up warning; "enter it again" status; back up the key ring with the database – Jojo George
Risk	Server address abused for SSRF (R-TFAS-05)	Address validator; no redirects – Jojo George
Open issue	⚠️ To be confirmed: without `settings.ai.edit` the inputs stay editable and only **Save** is hidden; whether the screen should be fully read-only.	Jojo George
Open issue	⚠️ To be confirmed: a user with `settings.ai.test` but without `settings.ai.edit` cannot test while there are unsaved changes (the automatic save is refused).	Jojo George
Open issue	⚠️ To be confirmed: only the screen ensures one active server; whether the API should refuse several enabled providers.	Jojo George
Open issue	⚠️ To be confirmed: exact conditions for **Ready to use** versus **In use** on the provider card.	Jojo George

15. Traceability (requirements → components)

FRD item	Module / component	Section
FR-TFAS-01	`AiProviderCatalog` (three providers)	§3.2, D-TFAS-3
FR-TFAS-02	`AiProviderSettingsComponent`; status from `AiProviderConfigService`	§4.3
FR-TFAS-03	`AiProviderCatalog` built-in models; `AiProviderSeeder`; `AiModel`	§6, D-TFAS-9
FR-TFAS-04	`AiProviderSettingsComponent` (server/model linking)	§4.3
FR-TFAS-05	`AiProviderSettingsComponent`; `AiProviderConfigService` remembers added models	§6.3
FR-TFAS-06	`AiProviderSettingsComponent`; `AiProviderConfig` address	§6.1
FR-TFAS-07	Request validator (address and SSRF rule)	§9, D-TFAS-6

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FRD item	Module / component	Section
FR-TFAS-08	Request validator (key rules)	§4.3
FR-TFAS-09	`AiProviderConfigService` (no key for Ollama)	§4.3
FR-TFAS-10	`AiKeyProtector`; write-only response	§9, D-TFAS-1, D-TFAS-2
FR-TFAS-11	`AiProviderConfigService` (key set, last four)	D-TFAS-2
FR-TFAS-12	`AiProviderConfigService` (empty key keeps the stored one)	§6.3
FR-TFAS-13	`AiProviderSettingsComponent`; validator (not both)	§4.3
FR-TFAS-14	`AiProviderSettingsComponent` (one server on); `AiProviderConfigService` save	§8.1, D-TFAS-7
FR-TFAS-15	Request validator (each provider once)	§4.3
FR-TFAS-16	`AiProviderConfigService` configured providers → TFA Assistant	§5.1
FR-TFAS-17	`AiProviderConnectionTester` → `ChatModelGateway` → provider clients	§8.2, D-TFAS-4
FR-TFAS-18	`AiProviderConnectionTester` (always a result)	D-TFAS-4
FR-TFAS-19	`AiProviderSettingsComponent` (save before test)	§14
FR-TFAS-20	`AiProvidersController` remove; `AiProviderConfigService`	§8.3
FR-TFAS-21	`AiProviderConfigService` removal rules	§8.3, D-TFAS-9
FR-TFAS-22	`AiProviderConfigService` key-readable check	§7
FR-TFAS-23	`AiProviderSettingsComponent` (cloud explanation)	§9
FR-TFAS-24	`AiProviderSeeder`	§6.3, D-TFAS-8
FR-TFAS-25	`AiProvidersController` per-action policies; component permission checks	§9
FR-TFAS-26	Permission catalogue prerequisites enforced by Role Configuration	§5.1, D-TFAS-5
FR-TFAS-27	Navigation builder and `permissionGuard`	§9
FR-TFAS-28	`AiProviderSettingsComponent` (toasts and error messages)	§8

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